

Assignment #4

General instructions:

- **Make sure to read all the exercises instructions** and to follow the instructions step by step by the order that I wrote.
- Make sure to all open the links that are in the assignment and read them.
- Do not wait to work on this assignment, start working on it whenever you can as it might take more time than you think.
- I work on these assignments a lot just for you guys and this is probably the most practical course you will have in the university, so I highly recommend that you really devote your time and effort and do your best to do them properly. It will benefit you a lot.
- If for some reason you weren't in class, please go over the slides and videos before starting this assignment.
- Solve the exercises by yourself. You can consult with friends when needed but **don't share your code with your friends**. Of course copying from others is unethical and will be reported.
- Solve the exercises using your self defined 'baby steps'. Don't start with a program that does everything. You can always create another small project in vs code just to make sure something works and then get back to your 'regular' project.

For each coding exercise, **before you start the exercise**, please create a **private** github repository just for that exercise **and invite me to that repository** (my github user is dk8827). While you are working on the exercise, once in a while(let's say every 30 minutes or so) or after you make any non-trivial change in your code, you should **commit and push to github your changes**. This will help you a lot in understanding how you work, what you did wrong etc.

Every time before you commit code, go over it(check all the files that changed and the diff of the content itself) and make sure that the commit makes sense.

Do not simply commit code that came out from an LLM. Sometimes LLMs do stupid things like removing your comments, deleting parts of your code, or adding some things that don't really make sense...

If you never worked with git and vscode, please watch/do an online tutorial like [this](#)

If it's a group exercise- make sure that all group members commit and push to that git repository, each one using his own github user. **Not committing and pushing to git properly will reduce your grade**. Also, your commit messages must be meaningful.

Here are some best practices for git commit messages:

<https://dev.to/sheraz4194/good-commit-vs-bad-commit-best-practices-for-git-1plc>

Exercise 1 - Multiple agents(freestyle)

Think of a cool idea for a project that involves using multiple agents. Completely freestyle and up to you. Start only when you have a cool idea (something that you would want to present in job interviews or show to your friends and family).

These are the only requirements:

- You should use pydanticAI for the agents. Read about pydanticAI and multi-agents here <https://pydantic.dev/docs/ai/overview/>
- You should use [Saperly](#) in some way.

(Reminder: Remember the 'wow' factor)

Exercise 2 - Bomberman ai competition

Download [this](#), read the docs, understand this, run things on your computer and when ready, submit your best prompts.

Reminder- let me know if you find a hack/ major bug so I can fix it.

Important- do not share your prompts with your friends.

Exercise 3 -

Now that you have the idea of your final project, you need to check the competition for it. Use LLMs to make 'Deep research' and find similar products/projects. After finding the products/projects, check out and try those products/projects yourself(if possible). Make a list of the things you want to add/do better/do differently. Add your project here and fill in all the columns

<https://docs.google.com/spreadsheets/d/11MI6KS-3K9Hvlzep60mYHA9nD6c-xzxr8YXAg1yknHQ/edit?usp=sharing>

Now you will implement the first(and most important) part of the project- the proof of concept. Think about what is the most critical part of your project that you are not sure on how to do it. The point is that this thing, when done, will prove to you that your final project is feasible.

If you develop something and then you realize it was too trivial, go on and implement another critical feature for your project.

Create a **2 minute** video that explains your project idea and shows your working proof of concept.

(Reminder: Remember the 'wow' factor)

You can create that video using any tool that you wish. This video will be shown in the final lesson to the whole class and we will be having a competition of the best video.

The submission of this exercise should only be a screen recording of you playing with your app/website's POC.

Submission instructions:

- Zip the code of exercise1 into a file called assignment3-exercise1.zip
- Zip the code of exercise3 into a file called assignment3-exercise3.zip
- Upload these zip files to Moodle. (if you see that your zip is large do not include .venv or other large directories that are not needed for the project to run).
- Make a high quality video screen recording of exercise1 running properly with all its features. Make sure that you go over all the important features. Use any screen recording software for that. Make sure that the video looks ok, understandable and the quality isn't bad. There are many such free recorders for windows and for mac.
- Make also a video of exercise 3 running properly with all the features.
- Upload each video file to google drive and create links that are **open to anyone with the link**
- Fill in [this](#) google form with the information about your exercises and the links to the videos.

Of course - filling the form with incorrect information is not allowed(for example, saying that everything runs fine while it doesn't) and will have a major impact on your grade.